

NPRB 2301: 2025 PI Meeting

Contact Information

Cindy cell: 907-723-0572

Santa Cruz NMFS Lab main line: (831) 420-3900

Meeting Details

February 26 – 28, 2025

Google Meet joining info

Video call link: <https://meet.google.com/snr-whcn-qdo>

Or dial: (US) +1 319-332-7230 PIN: 821 565 258#

More phone numbers: <https://tel.meet/snr-whcn-qdo?pin=2494536584035>

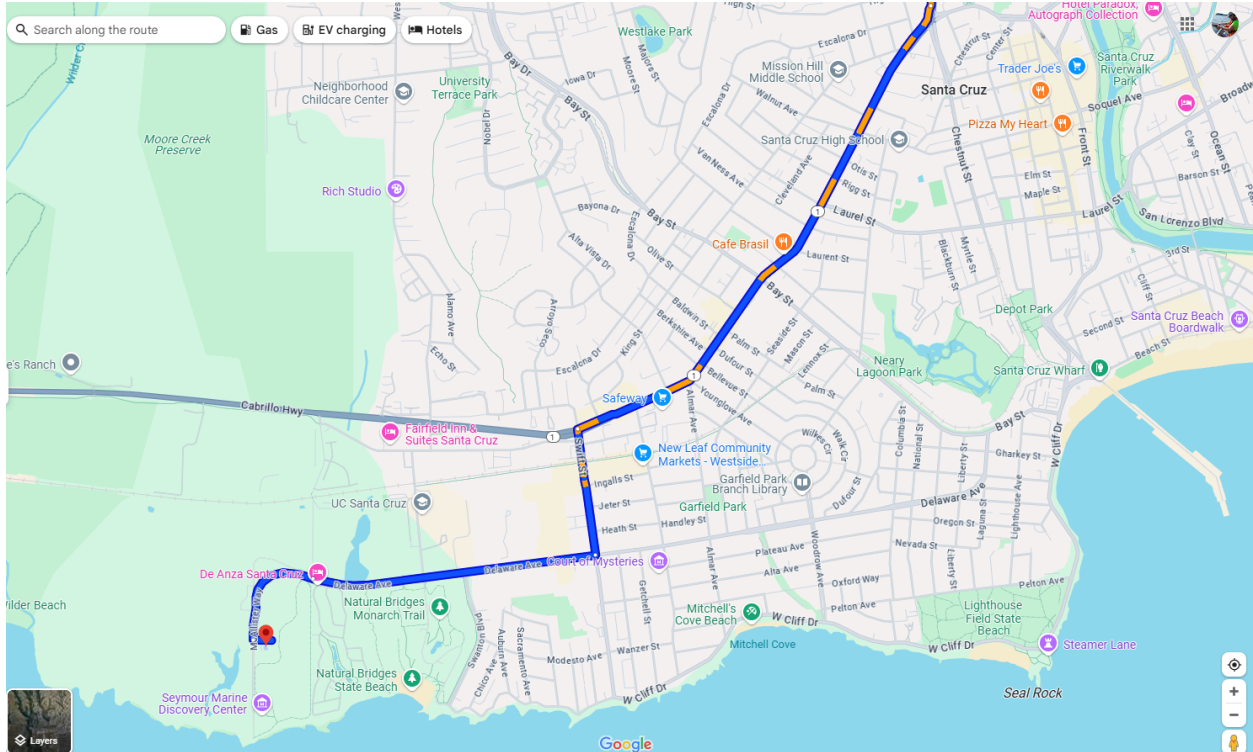
Slide link:

https://docs.google.com/presentation/d/1LlihplMjEbHRWzHpIAJSysJrwV1dGEWScetABpVyPPw/edit#slide=id.g3318636d288_0_391

Meeting Logistics

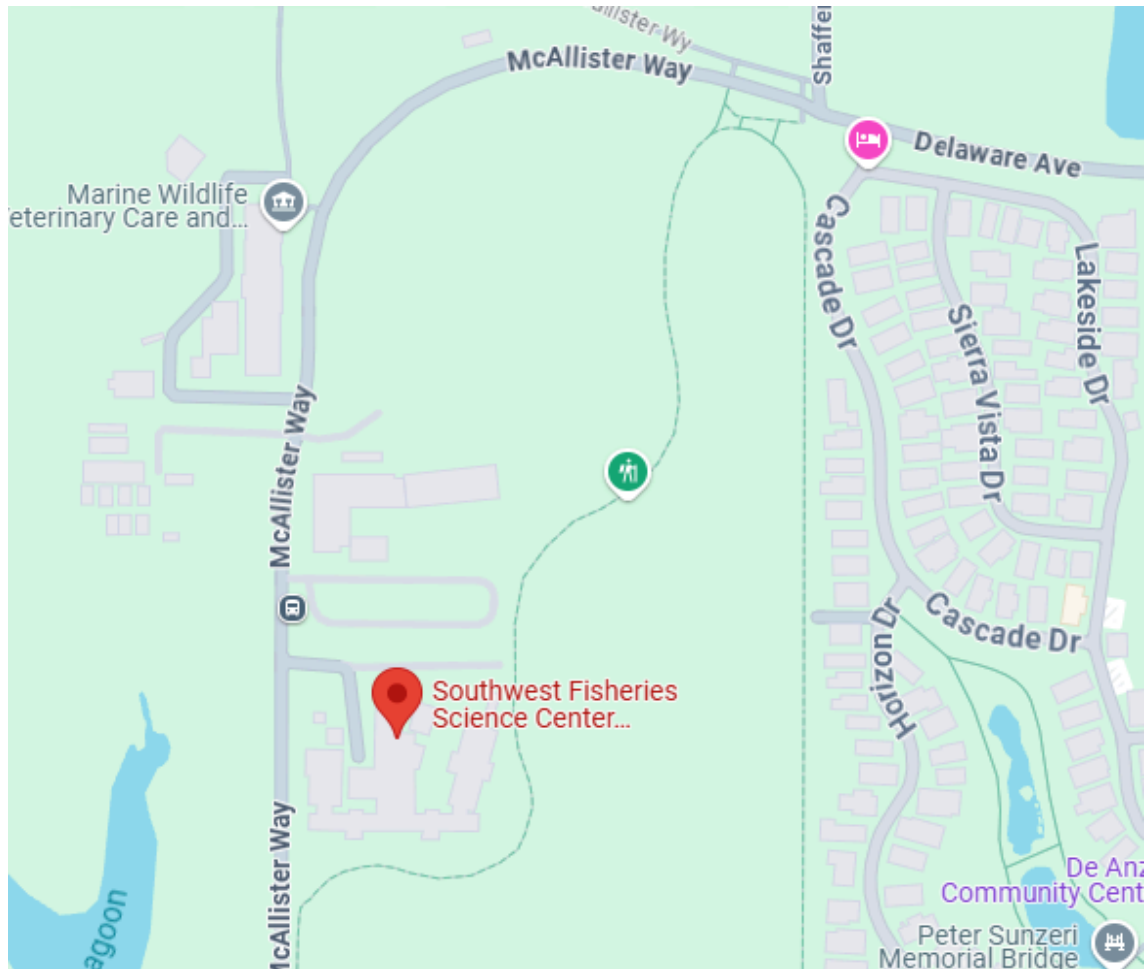
Location

Hosted at the [NMFS Santa Cruz Laboratory](#) Conference Room 209 (aka 'The Fishbowl')
110 McAllister Way, Santa Cruz, CA 95060



Building Access and Parking

The closest free parking is on Delaware Avenue, and a short (0.5 mile) walk through what looks like a lovely trail. There is paid parking (\$10/day) in the first parking lot on the left after you enter the UC Coastal Science Campus, Lot 206. Instructions can be found [here](#). If you plan to drive in, please contact Cindy for arrangements to reimburse parking and mileage.



The building is a locked facility. Text Cindy (907-723-0572) when you arrive, so we can let you in. There is also a phone at the entryway to call to be let in. All visitors will need to sign into the Visitors Log and wear a visitors label with both first and last name.

Lunch and Refreshments

We were recommended to bring in our lunches, as it can take time to drive to the nearest locations. There is a fridge we can use to store food. There is also a coffee maker. I plan to bring some light snacks!

Onsite Internet Access

Details forthcoming

Agenda

This agenda is intended to maximize participants' availability, thus some topics are out of order. Please let Cindy know if you have concerns about timing of any topics.

Date	Time	Topic	Lead
Wednesday, Feb 26	9:00 - 10:00	Getting things set up/ troubleshooting tech issues	Cindy
	10:00 - 12:00	Objective 3/ M estimation	Cindy
	12:00 - 1:00	Lunch	
	1:00 - 1:30	Review project objectives, manuscript plans, etc	Cindy
	1:30 - 5:00	14C results and planning	All
Thursday, Feb 27	9:00 - 10:00	Work Session as needed (Cindy has a class conflict)	
	10:00 - 12:00	Work session	Cindy
	12:00 - 1:30	Lunch	
	1:30 - 3:00	Project updates/budgets/logistics	Cindy
	3:00 - 5:00	Dogfish samples	All
Friday, Feb 28	9:00 - 10:00	Work Session as needed (Cindy has a class conflict)	
	10:00 - 11:00	Eye delamination protocols	Cindy/Charlotte
	11:00 - 12:00	Soluble protein plan of attack	Cindy/Dan/Bruce
	12:00 - 1:00	Lunch	
	1:00 - 3:30	Stable Isotope (results and/or planning)	Taylor
	3:30 - 5:00	Revisit any questions/topics	All

Notes

Feel free to add specific questions or topics within a session.

Wednesday Feb 26, 2025

Objective 3 - plausible M estimates

- Use approach similar to Sullivan et al. to review incorporate all potential data sources
- Other ideas?
- Develop some framework for the analytics, including sensitivity analysis
- [Natural Mortality Tool](#) - worked through the tool during the meeting using some current plausible estimates of various life history parameters. Can use it to estimate a range of plausible M values.
- [Sullivan](#) framework (Cindy will add link to rockfish M methods spreadsheet). We can adapt Jane's code to estimate M. Use Cindy's method from GOA OROX assessment to weight M values to estimate single best value and uncertainty. End goal will be Tech Memo for all shark and skate species (Tribuzio and Matta)
- Propose doing a bonus fecundity analysis - estimate reasonable bounds
- Can make reasonable estimates of age at maturity based on Garrett Dunne's meta-analysis
- Two products expected:
 - Tech Memo for all sharks and skates - plausible estimates of M from all life history proxy methods
 - Manuscript - Sleeper shark specific M and fecundity

Review project objectives, manuscript plans, roles

[Project Logistics document](#)

[Roles and manuscripts plan document](#) (also objectives)

In-Meeting Notes

- By end of the week, want to define manuscript goals for Objectives 1 and 2 (following group discussions)
- Goal is to have all samples shipped to WHOI/Stanford/UCSC by Aug 31, 2025
- Stanford's billing process is fairly straightforward, should be pretty fast going forward.
- CSIAA will be run at UCSC, still need to decide on what samples (decide this week)
- WHOI's turn-around is slower than the other labs due to extenuating circumstances
-

14C Results

- [Layer results](#) spreadsheet (locked)
- [Layer data spreadsheet](#) is active (can add data to)
- Preliminary results (Allen)
- How do things look? Any changes needed?

- Future planning
 - Priority samples/layers
 - Do we want to run all layers for all animals, hold off on “remainder” layer, subsample, etc?
 - Are sequential lens measurements (from photos) needed for every eye?

In-Meeting Notes

- Use 420 as an example for reasonable sampling resolution
- Strange pattern in specimen 306 - steep decline, seems to indicate it's older
- We need more total lens diameter measurements to generate a predictive relationship to back-calculate fish size at diameter.
- Do comparison of left and right, then go consistently with one side going forward (goal is one eye per fish). Prioritize ranks 3 and 4 eyes first.
- Length info is unknown type for some fish (PCL or TL), can't determine which. Layer diameters might be able to help identify. Or, consider the variance. May be within the range of measurement error anyway.
- May be difficult to fill length gaps. Should we prioritize certain length bins for C14? Small sharks are showing highly variable post-peak values. The most diagnostic sharks will be in the 3-4 m range. High sample resolution for these is very important. Also, prioritize dogfish heads (one eye per animal), embryos (also measure embryo length), and candle yolk.
- Possibility of Beth and Lisa coming up to Juneau to process eyes (another 3-day workshop - July ???)
- Do we need to keep saving the remainder layer? It is very hard to get in and out of the vials. Charlotte nomenclature - Capsule is full lens (layer 1). Remainder is the aqueous humor (clear liquid), (labeled as layer 2 in the layer data) between the core and the capsule. Capsule should be outermost. The batch we just submitted should have both capsule and remainder. Check to see how those values differ and decide what to do. (Note, there is some divergence in our terminology in our earlier work, in terms what we called the remainder.) We are not using the word remainder any more - it's now layer 2.

Thursday Feb 27, 2025

Data workflow and data management

- Monthly project reports
- Questions about uncertain lengths

Project Updates/Budgets/Logistics/etc

- Planning for uncertain times
 - What if gov shuts down March 14th
 - Trying to process as many frozen eyes as possible to get them out of ABL

- Can we attempt to do the protein separation as well?
- Move processing to UAF during shutdown
 - Need to confirm lab space
 - Freezer access?
 - Can Fletcher grab stuff for us from the lab during shutdown?
 - Charlotte will organize samples in boxes to transfer to UAF: one eye of pairs for ranks $\frac{3}{4}$ >300 cm.
 - Pack supplies and moving plans by March 14th, with the plan to move supplies and equipment during the 4 hr orderly shut down on the 17th
- Send all samples for AMS/SI to Taylor on Mar 14th
 - Layers that were recently peeled
 - New dogfish
 - Some of the “contaminated” samples that have pairs either already with 14C or that we have frozen eyes for
- Longer term planning
 - Issues with ABL accessing funds due to end of FY (Sept 30) - assuming no more AFSC expenditures after this Sept 30
 - Travel to AMSS Jan 2026?? Anyone who still has money to spend out in their budget can join
 - Alternative plan to maximize the use of our funds
 - Send left over AFSC funds to AKBiomap for Charlotte or WHOI for samples - depends on what funds are left - plan is to evaluate funds in July and make plans
 - Can WHOI bill early for samples so we can get the budgets spent out?
Not likely
- Bi-annual reporting

Dogfish Samples

- Spines cleaned, aged by Beth
- Archive of UAF and ADFG spines, are they useful for anything
- How to process embryos?
- No eyes done yet
- Lisa Hillier (WDFW) working on parallel project and will try to provide some dogfish
- Charlotte will begin delamination of remaining eyes, but only one of pair will be shipped for AMS
- Whole eye lenses will be dissected out for embryos, no peeling. Candles: scoop of yolk taken out and dried. Charlotte will try to take length measure of embryos to give an indication of gestation age
- Lisa H also looking in archived samples at WDFW to see if there are any that may have birth year in the target zone
 - We need to examine our archived spines for this as well, could some of the 1990 ADFG samples?

- DFO has samples from 1984 and 1989 that they gave to WDFW but Lisa would need permission to use, WDFW's samples start in 2004. Cindy has some from the 1990s (Cook Inlet or SEAK).

Friday Feb 28, 2025

Eye delamination protocols

Review [document](#) and discuss concerns.

In-Meeting Notes

- Document is still in progress and is being updated. Charlotte has handwritten notes that she will add.
- Bruce suggests doing a swipe test for the lab that Charlotte will be using in the event of a shutdown.
- Don't need to send remainder layers for AMS
- For dogfish embryos, some mothers have 2 pups. What is the best way to prioritize those. Bruce - one eye should be enough. Cindy - there's a mixture of near-term and halfway developed embryos. Will also do the candle yolk.
- One sample (543) from one of our largest sharks that also had maturity info indicating maturing or mature stage, had a very different lens structure with what looked like an additional capsule, and more gelatinous material. May be related to maturity? Dan saw something similar with the dogfish samples. This information might also help corroborate maturity stage for sleeper shark (Garrett's work). Dan sees this in humans as well.
- Dan has noticed that cooler lenses are a little easier to work with. Charlotte has been sometimes using a little drop of DI water to help release the layers to make them easier to peel.
- Try an icebath method for the dissection dishes to keep the eye cool, may make it more viscous, keeping it from solidifying. Alternatively could use small amounts of DI water to re-soften the core.

Soluble vs insoluble protein

- Lab testing of separations
- Selection of samples to move forward with

In-meeting notes

- We are often at the minimum mass for layers. Would there be enough mass? 2 mg is good for stable isotopes

- Most of the mass is likely to be in the soluble. For humans, most mass is transparent crystalline proteins. Insoluble were much much smaller. Shark eyes will probably be the same.
- Insoluble protein gives the birthdate of the cell. Determine whether there's turnover/repair. Soluble was younger in humans.
- Use second eye from fish that have had AMS done and have length data
- Do multiple layers per eye. Probably not going to see as much difference for young fish as for old fish.
- Sample size is 5 fish per species, 4 layers per fish (from proposal) for 40 total but we can adjust this. Layers probably don't need to be as fine as for the C14 age analysis (and because insoluble portion is so small, may have mass limitations).
- Centrifuge speed - Dan used 20,000 G. Bruce thinks half that speed or less will probably work. The portions will eventually separate out in water, centrifuging just helps it happen faster. A pellet will form for insoluble proteins. Cindy will follow up with Matt and Emily about which equipment we can use. Can use glass Pasteur pipettes (baked).
- One issue with the soluble and insoluble portions after spinning is that you can end up with a powder inside the vial that is difficult to recover. Bruce suggests freezing it with a little bit of water, freeze it, and ship to Bruce on dry ice. Taylor doesn't have a way to deal with specimens in liquid but NOSAMS can preprocess and can probably handle samples that are in small volumes of liquid (charge a higher rate).
- Taylor - may be able to use tin EA capsules (tend to have low C). Silver EA capsules that are baked in the muffle furnace could also work. Silver would be preferable to tin. Could be used for both soluble and insoluble. Put in drying oven after separating, then fold up, put in an EA sample tray, 96 micropipette tray, or GC vials, and ship to Taylor. Taylor can give more guidance (in person or virtual) on how to do this.
- Charlotte and Cindy will test these methods to determine whether the sample is powder-like and hard to recover or not. (Try first with length-less eyes.) UAF may have facilities that can be used.
- Goal is to do this portion of the project after shutdown if one occurs.

Stable Isotope

- Any early results?
- Should we plan to run isotopes on all layers that we do AMS on (if sufficient sample is available)?
 - What are decision points to prioritize Stable Isotope samples?
- CSIAA stuff.....nothing yet, how do we select layers for that?

In-Meeting Notes

- Layers can be combined if need be to achieve adequate sample mass
- Outputs are delta C13 and delta N15 (with uncertainties)

- Cindy will add those outputs to the layer results spreadsheet
- Nitrogen isotopes = trophic indicator (become enriched with increasing trophic level). Ontogenetic changes. Need to know N15 at base of food web (can be affected by migration) Can be difficult if they migrate large distances. CSIAA will help with this - some amino acids don't enrich and match the food web base.
- Carbon isotopes - can tell food web base source (enrich but not as much as N)
- Taylor is planning on consulting with some other experts on interpretations (his PhD adviser might have a student who is interested in participating)
- Matt Rogers might also be helpful
- Planning to analyze eyes of dogfish embryos and moms for stable isotopes
- Contamination is less of an issue here (especially for N) - still value in sending the suspected contaminated tissues.
- Can shift more budget towards stable isotopes (relatively inexpensive)
- Will ship what is available now for stable isotopes. Can do rest after potential shutdown.
- CSIAA is more expensive and a lot more labor. Only doing this on PSS. Need 2 mg for sample mass. We budgeted for 20 samples. Inner vs outer comparison would probably be most useful. Would give info about living in areas with different food web bases and also ontogenetic trophic shifts. Could also do sequential sampling in one larger fish. Could shoot for 5 layers each from 2 biggest fish (the big AI one and the big WC one). Do inner and outer layers from 5 additional fish that are large and have 2 eyes and good ancillary data. This will happen after shutdown.
- Taylor could use a dessicator to store vials (add to shopping list). Can ship back to Cindy after we get AMS and stable isotope results for storage until papers are published.
- Clamshell or cardboard boxes for vials would be helpful for shipping and storage (Cindy can send some to Taylor)

Action Items

- ☐ Copy google documents (e.g., roles/logistics docs) to github for easier access
- ☐ Add link to rockfish M estimates and methods spreadsheets
- ☐ Add species column to layer results spreadsheet
- ☐ Take measurements of all eye lens diameters (at least whole lenses to start)
- ☐ Send out poll to figure out the best time for another 3-day workshop (tentatively week of July 7th)
- ☐ Determine leads and subjects of manuscripts for Objectives 1 and 2
- ☐ Check github page and make sure all can access and that readme is OK
- ☐ **EVERYONE:** Review most recent NPRB2301_YYYY_MM_DD.pdf and provide feedback to Cindy
- ☐ Shopping list pre shutdown.

- ☒ ~~Check on what happens with shipping in the case of a shutdown~~
- ☐ Investigate alternative locations for eye dissections and freezer space in case of shutdown- UAF is the first choice. Can coordinate box transfer with Fletcher Sewell. Organize eye boxes (Chartotte)
- ☐ Ask Matt and Emily about centrifuge for insoluble vs soluble protein comparison
- ☐ Do swipes of UAF lab
- ☐ Review otolith CSIAA papers (Taylor and Beth)